# RESEARCH STATEMENT – ABHIJIT S. MUDIGONDA

## 1. Introduction

1.1. **The Langlands programme.** A recurring theme in the Langlands programme of algebraic number theory is the interplay between Galois representations[1], automorphic forms,[2] and algebraic varieties.[3]

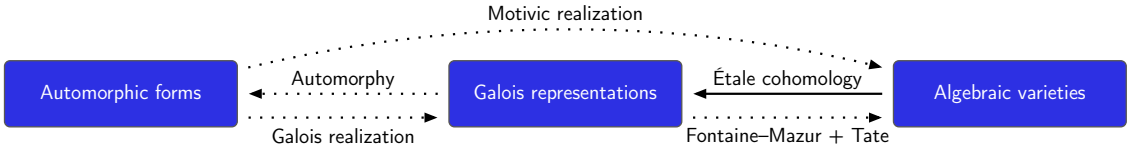
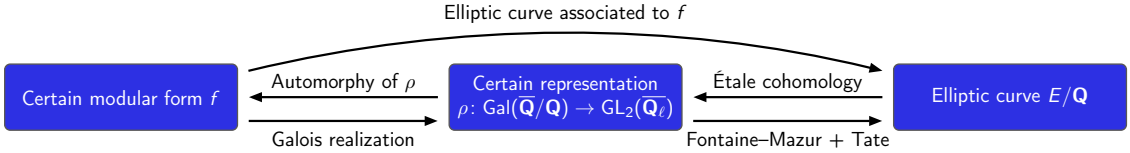



Figure 1. A rough picture of some correspondences predicted by the Langlands programme

In general, the correspondences of Figure 1 remain conjectural, and progress, even in special cases, has been hard-won. Perhaps the most famous result in algebraic number theory – the 1994 proof of Fermat's last theorem due to Wiles [Wil95] and Taylor–Wiles [TW95] – proceeds by exhibiting the automorphy of (the Galois representations associated to) most elliptic curves over $\mathbf{Q}$. The ideas of Wiles and Taylor–Wiles were later extended to all elliptic curves over $\mathbf{Q}$ by Breuil–Conrad–Diamond–Taylor [BCDT01]. Combined with work of Shimura [Shi71], this completes Figure 1 for elliptic curves over $\mathbf{Q}$. For elliptic curves, the relevant automorphic forms in Figure 1 are a certain subset of a class of functions called modular forms.




Understanding the correspondences of the Langlands programme for interesting classes of algebraic varieties furnishes us with new angles with which to study these varieties. For example, the analytic continuation of the $L$-function associated to an elliptic curve $E/\mathbf{Q}$ – necessary to even formulate the Birch and Swinnerton-Dyer conjecture – is difficult to prove directly, but follows from the (much easier to prove) analytic continuation of the $L$-function of the modular form $f$ associated to $E$.

Elliptic curves are the 1-dimensional examples of an important class of algebraic varieties called abelian varieties. **It is a major open problem in algebraic number theory to understand Figure 1 for higher-dimensional abelian varieties**. Recent work of Boxer–Calegari–Gee–Pilloni [BCGP25] proves the automorphy of most abelian surfaces (2-dimensional abelian varieties) over $\mathbf{Q}$. Combined with a construction of Taylor [Tay91], we obtain many, but not all, of the arrows of Figure 1 for abelian surfaces. When $g \geq 3$, proving even the automorphy of generic $g$-dimensional abelian varieties seems completely out of reach of current methods. **However, if we restrict to higher dimensional abelian varieties which are "atypical" in an appropriate sense, we can hope to make progress by exploiting extra structure that is available in these cases**. Of particular interest is the case $g = 4$ (henceforth, "**abelian fourfolds**"), as four is the smallest dimension in which atypical abelian varieties exhibiting certain exotic phenomena exist. As with elliptic curves, understanding the correspondences of the Langlands programme for interesting classes of abelian varieties gives us new avenues towards understanding these varieties.

[1] Representations of the absolute Galois group $G_K := \mathrm{Gal}(\overline{K}/K)$ on an $n$-dimensional vector space.
[2] Roughly, analytic functions on complex manifolds satisfying symmetries with respect to a discrete subgroup of $\mathrm{GL}_n(K)$.
[3] Or rather, $n$-dimensional pieces of the cohomology of algebraic varieties over $K$.

1.2. **The role of computation.** From the numerical plots that helped motivate the Birch and Swinnerton-Dyer conjecture [BSD65] to the tables of modular forms in the LMFDB [LMF26], examples and explicit computation have historically played an outsize role in algebraic number theory. The Langlands programme is no different. Anecdotally [Ser02], once Weil formulated a conjecture about the precise relationship between an elliptic curve and its associated modular form, a bevy of computational evidence was gathered to reinforce and refine the conjecture. The same principle holds true today. **Explicit computations of automorphic forms on the one hand and geometric objects on the other motivate the statements of new automorphy theorems and provide insight towards their proofs**.

1.3. **Disclosure of AI use.** AI was not used in the ideation nor in the writing process of this statement beyond reference lookups, LaTeX formatting assistance, and very light proofreading. At the end however, I used GPT-6 Astra to confirm that my proposed future work is neither immediately rejected by nor immediately trivialized by current publicly available models. While I am not inherently opposed to the use of AI in mathematics, the graduate work I describe was almost entirely completed prior to the recent increases in capability of LLMs.

## 2. Graduate work

My graduate work to date can be split into two streams, one theoretical and one experimental, each feeding into the other. **On the theoretical side, I study atypical abelian fourfolds and their relationship to automorphic forms**. **On the experimental side, I develop and apply algorithms for computing spaces of Hilbert modular forms** (a special type of automorphic form). In the following two subsections, I discuss each of these streams, and their relationship to the work I hope to do going forward.

2.1. **Potential automorphy of atypical abelian fourfolds.** As was discussed in Section 1.1, automorphy of 1- and (most) 2-dimensional abelian varieties is known, but current methods appear insufficient to prove automorphy of generic $g$-dimensional abelian varieties when $g \geq 3$. However, long before automorphy of generic 1- and 2-dimensional abelian varieties was known, progress was made on the automorphy of atypical abelian varieties of these dimensions. Roughly, an abelian variety $A/\mathbf{Q}$ is atypical if it possesses some extra geometric structure that a generic abelian variety does not.[4] Any abelian variety has endomorphisms given by multiplication by $n \in \mathbf{Z}$. For a typical abelian variety $A$ this is all there is, and $\mathrm{End}_{\overline{\mathbf{Q}}}(A) = \mathbf{Z}$. A major class of atypical abelian varieties consists of those with $\mathrm{End}_{\overline{\mathbf{Q}}}(A) \supsetneq \mathbf{Z}$. **In** [Mud26a]**, I prove the potential automorphy** (a weaker notion of automorphy) **of many atypical abelian fourfolds over Q.** For a geometrically simple abelian fourfold, a classification of Albert [Alb39] shows that $\mathrm{End}^0_{\overline{\mathbf{Q}}}(A) := \mathrm{End}_{\overline{\mathbf{Q}}}(A) \otimes \mathbf{Q}$ is either a totally real field, a division quaternion algebra, or a CM field. A finer stratification is given by the (geometric Albert) "type" of $A$ (see e.g. [MZ95]). In Table 1, I describe my potential automorphy results in terms of the type of $A$.

| $\mathrm{End}^0_{\overline{\mathbf{Q}}}(A)$ | Totally real field | | Quaternion algebra | | | CM field | | |
|---|---|---|---|---|---|---|---|---|
| Type | I(2) | I(4) | II(1) | II(2) | III | IV(1,1) | IV(1,2) | IV(1,4) |
| Status | | | | | | | | |

Table 1. A summary of the status of potential automorphy of atypical abelian fourfolds over $\mathbf{Q}$ after [Mud26a]. "Type" is shorthand for geometric Albert type. Solid green indicates potential automorphy, hatched yellow indicates potential automorphy in some but not all cases, and dotted red indicates that potential automorphy remains open. Prior to our work, all cases except for that of Type IV(1,4) were open.

At a high level, my approach to proving the potential automorphy of an atypical abelian fourfold $A/\mathbf{Q}$ proceeds in three steps. Let $\rho_A$ be the Galois representation associated to $A$.

[4] Precisely, $A$ is atypical if the traceless Lie algebra of its Mumford–Tate group is strictly contained in $\mathfrak{sp}_{2g}$.

(1) **Prove a structure theorem for** $\rho_A\colon G_{\mathbf{Q}} \to \mathrm{GL}_{2g}(\overline{\mathbf{Z}_\ell})$ ([Mud26a, Main Theorem 1]) and apply existing results from the Langlands programme to reduce the potential automorphy of $\rho_A$ to the potential automorphy of a representation $\xi\colon G_K \to \mathrm{GL}_m(\overline{\mathbf{Z}_\ell})$ for some number field $K/\mathbf{Q}$ and some $m < 2g$. Intuitively, we reduce the dimension of our representation at the cost of extending our ground field.
(2) **Obtain an automorphy lifting theorem**: roughly, that $\xi$ is automorphic if (along with some additional hypotheses) the mod-$\ell$ reduction $\overline{\xi}\colon G_K \to \mathrm{GL}_m(\overline{\mathbf{F}_\ell})$ is automorphic in an appropriate sense.[5]
(3) **Construct a moduli space of abelian varieties whose points act as "bridges" between $\xi$ and a representation $\xi'$ which is known to be automorphic.** A Moret–Bailly argument [MB89] then shows that this moduli space has points over an appropriate field.

In [Mud26a], I first prove[6] the structure theorem mentioned in step 1. In step 2, I apply the existing automorphy lifting theorems of [BCGP21] and [ACC+23]. In step 3, to deal with subtleties involving the field of definition of the geometric endomorphisms of $A$, I construct certain twists of standard moduli schemes that capture abelian varieties with a prescribed Galois action on their endomorphism algebras, and run the Moret–Bailly argument on these schemes.

**In work in progress, I am addressing the potential automorphy of abelian fourfolds of (geometric Albert) type III.** In the cases treated in [Mud26a], I applied existing automorphy lifting theorems to the $\xi$ appearing in step (1). However, the representations $\xi$ appearing in this case are four-dimensional orthogonal representations with Hodge–Tate weights $(0,0,1,1)$, and an automorphy lifting theorem in this setting is not known. I expect these Galois representations to be associated to the tensor induction (an analogue of the induced representation construction with tensor products replacing direct sums) of a 2-dimensional representation associated to a Hilbert modular form $f$ (see Section 2.2) of weight $k = (1,2)$. To prove this, I am using Hida theory arguments in the style of Buzzard–Taylor [BT99], and Buzzard [Buz03]. A particularly interesting feature in this setting is that, because the entries of $k$ are not all congruent modulo 2, $f$ is non-algebraic (as an automorphic form on $\mathrm{GL}_{2,F}$). Non-algebraic automorphic forms are not well studied, as they do not fit nicely into the picture of Figure 1, and are not expected to be directly associated to geometry. Here however, **even though $f$ itself is nonalgebraic, a tensor construction associated to $f$ is algebraic and is expected to be associated to geometry.**

2.2. **Computing Hilbert modular forms.** Let $\mathcal{H}$ denote the complex upper half-plane, let $F/\mathbf{Q}$ be a totally real field of degree $n$, and let $\mathbf{Z}_F$ be the ring of integers of $F$. The basic example to keep in mind is when $F$ is a real quadratic field and $n = 2$. **A Hilbert modular form $f$ over $F$ is (roughly) a holomorphic function on $\mathcal{H}^n$ which is invariant under an action of a discrete subgroup of $\mathrm{GL}_2(F)$.** This discrete subgroup is specified by an ideal $\mathfrak{N} \subseteq \mathbf{Z}_F$ called the level, and the action is specified by a tuple $k \in \mathbf{Z}^n_{\geq 1}$ called the weight and a character $\chi\colon (\mathbf{Z}_F/\mathfrak{N})^\times \to \mathbf{C}^\times$ called the nebentypus. For a given level $\mathfrak{N}$, weight $k$, and nebentypus $\chi$, the space of Hilbert modular forms with these parameters is a finite-dimensional vector space which we denote by $M_k(\mathfrak{N}, \chi)$.

While Hilbert modular forms are a priori analytic objects, certain Hilbert modular forms – called Hilbert eigenforms – also carry algebraic information, and Hilbert eigenforms with interesting properties often correspond to geometrically interesting objects via the correspondences of Figure 1. **A key feature of Hilbert modular forms in this context is that they are amenable to explicit computation**, and one can hope to search for interesting objects by first finding a corresponding interesting Hilbert modular form, and then using the form to understand or reconstruct the object. For example, we discussed in Section 2.1 how certain weight $(1,2)$ Hilbert modular forms are conjecturally associated to abelian fourfolds of type III.

A Hilbert modular form can be specified by a power series in $n$ variables called its $q$-expansion. We say that a $q$-expansion is specified to precision $B$ if we are given the coefficients of the power series for terms of "size" at most $B$, for a suitable notion of size.

[5] Specifically, if $\overline{\xi}$ is the residual representation of the Galois representation associated to some automorphic form on $\mathrm{GL}_{m,K}$.

[6] A weaker version of this theorem was initially proved by [FG22] using similar techniques, but we were unaware of their work until very recently.

**Problem 2.1.** *Given a precision $B \in \mathbb{N}$, a field $F/\mathbf{Q}$, and a level $\mathfrak{N}$, weight $k$, and nebentypus $\chi$ as defined above, return a sequence of $q$-expansions, each specified to precision $B$, which span $M_k(\mathfrak{N}, \chi)$.*

Prior to 2022, algorithms for Problem 2.1 required that $[F : \mathbf{Q}] = 2$, $\chi = 1$, and $k \in \mathbf{Z}^n_{\geq 2}$. Additionally, these algorithms required that $k$ be paritious, meaning that the entries of $k$ all be congruent modulo 2. In a series of papers, one with several coauthors, one joint with Eran Assaf, and one solo, **we were able to fully resolve Problem 2.1 for paritious weights, and also for nonparitious weights subject to a technical condition.**

**Theorem 2.2** ([ABB+26, AM26, Mud26b], building on [DV13])**.** *Let $F/\mathbf{Q}$ be a totally real field of degree $n$. There is an explicit algorithm which solves Problem 2.1 for any level $\mathfrak{N} \subseteq \mathbf{Z}_F$, paritious weight $k \in \mathbf{Z}^n_{\geq 1}$, and arbitrary nebentypus character $\chi$. When $F$ has narrow class number 1, we may remove the assumption of paritious weight.*

A key primitive in Theorem 2.2 is multiplication of $q$-expansions of HMFs. For example, multiplication is an essential step in the algorithm of Theorem 2.2 for computing spaces of Hilbert modular forms of weight $k \in (\mathbf{Z}^n_{\geq 1} \setminus \mathbf{Z}^n_{\geq 2})$, like those expected to be associated to fourfolds of Mumford type. The $q$-expansion of a Hilbert modular form given to finite precision generally has infinitely many terms, but by exploiting redundancy between the coefficients, **my joint work gives an algorithm which multiplies Hilbert modular forms of precision $B$ in time $O_F(B \log B)$** (where the subscript indicates that constant in the big-$O$ depends on $F$), which is optimal up to log-factors.

**In [ABB+26], we also give a uniform approach to access the birational geometry of Hilbert modular varieties.** Any Hilbert modular variety is birational to a projective variety determined entirely by a certain finitely generated graded ring of Hilbert modular forms. **We give an algorithm to provably compute this graded ring**, using the Eichler–Selberg trace formula [Hij74] and a Jacobian criterion for when we have found a generating set. Previously, computing a model of any given such variety was a paper unto itself, requiring bespoke geometric arguments (see e.g. [Gun63, Hir77, Res74, vdGZ77]). **With our algorithm, we are able to recover the results of these papers with the push of a button.**

In order to better understand the type III abelian fourfolds discussed in Section 2.1, I am now applying Theorem 2.2 to search for Hilbert modular forms of weight $(1, 2)$. These forms always have nontrivial nebentypus, so essentially the full strength of Theorem 2.2 is needed to search for these objects. Explicit examples of type III abelian fourfolds were constructed in [CFLV23]. **By searching for the corresponding weight $(1, 2)$ Hilbert modular form, I hope to explicitly realize the automorphy of such an example.**

The implementation the algorithms of Theorem 2.2 is part of an open source repository for producing and manipulating Hilbert modular forms. Other researchers have already been able to use code that I have written in their own work [Log26, Log25]. I plan to soon incorporate tables of Hilbert modular forms computed using Theorem 2.2 into the LMFDB [LMF26], a widely used database of arithmetic objects.

## 3. Proposed future work

3.1. **Abelian fourfolds of Mumford type.** In [Mum69], Mumford constructed families of abelian fourfolds (eponymously referred to as "abelian fourfolds of Mumford type") with several unique properties. For example, they are the simplest examples of atypical abelian varieties with $\mathrm{End}_{\overline{\mathbf{Q}}}(A) = \mathbf{Z}$. **Another motivation for studying these fourfolds is that they are the simplest abelian varieties over $\mathbf{Q}$ for which the Mumford–Tate conjecture** – a question closely related to the Hodge conjecture – **remains intransigent.**[7] In the context of the Langlands programme, the Mumford–Tate conjecture relates the middle and right pieces of Figure 1. Similar conjectures have been proven for some varieties by using automorphic forms associated to the variety as a bridge between the variety and its Galois representation [Rib80, Mom81, MR87, LZ22].

In [Mud26a], I prove potential automorphy for several classes of atypical abelian fourfolds, but do not address fourfolds of Mumford type. However, like the case of abelian fourfolds of (geometric Albert) type III discussed at the end of Section 2.1, the Galois representation associated to an abelian fourfold of Mumford

[7] Given an abelian fourfold whose Galois representation "looks like" that of an abelian fourfold of Mumford type, it is not known how to prove that it actually has the extra geometric structure characterizing fourfolds of Mumford type.

type is expected to be the tensor induction of a 2-dimensional representation associated to a nonalgebraic Hilbert modular form [Cal21].

3.2. **The Langlands programme for abelian fourfolds of Mumford type.** I propose to study aspects of the Langlands programme in the context of abelian fourfolds of Mumford type. I hope to leverage this understanding to attack the Mumford–Tate conjecture for abelian fourfolds. This setting introduces many unique challenges, reflecting the unusual properties of these fourfolds. I discuss some of the specifics in Section 3.3, but at a high level I have three initial goals.

(1) Use generalizations of the arguments of Buzzard–Taylor [BT99] and Buzzard [Buz03] coupled with ideas of [Pat19] and geometric work of [Gal00] to **deduce the automorphy of abelian fourfolds over $\mathbf{Q}$ of Mumford type**. My ongoing work on the potential automorphy of abelian fourfolds of type III (see Section 2.1) will serve as a starting point for this work.
(2) **Build on ideas of Oda [Oda81] to formulate conjectures about the periods** – complex numbers which are intimately related to the Mumford–Tate conjecture – of such fourfolds in terms of the relevant automorphic forms. Oda's conjectures are about an important class of algebraic surfaces called Hilbert modular surfaces. Similar to abelian fourfolds of Mumford type, the Galois representations associated to Hilbert modular surfaces are related to tensor inductions of 2-dimensional Galois representations associated to Hilbert modular forms. I expect that similar ideas will help us analyze the periods in our setting.
(3) Using explicit families of such fourfolds constructed in [BHPS25] and algorithms that I have helped develop [ABB+26, AM26, Mud26b] (see Section 2.2) for computing the conjecturally associated automorphic forms, **understand the automorphy and the periods of concrete examples of such fourfolds**.

Murty–Ramakrishnan [MR87] use period relations coming from automorphic forms to prove the Mumford–Tate conjecture for Hilbert modular surfaces. As discussed above, the Galois representations of Hilbert modular surfaces share certain features with the Galois representations of abelian fourfolds of Mumford type. Given automorphy of abelian fourfolds of Mumford type and an understanding of their periods in terms of automorphic forms, **I hope to attack the Mumford–Tate conjecture for abelian fourfolds with a similar strategy**.

3.3. **Some technical details.** In the remainder of this section, we outline a speculative strategy for proving the automorphy of (the Galois representation $\rho_A$) associated to) an abelian fourfold $A$ of Mumford type. Here, $\rho_A \cong \bigotimes\text{-}\operatorname{Ind}_{G_K}^{G_{\mathbf{Q}}} \xi$ for a totally real cubic field $K/\mathbf{Q}$ and a 2-dimensional representation $\xi$ which is not Hodge–Tate and does not fit into a compatible family of Galois representations. By a (fractional) Hodge–Tate weight computation, we expect $\xi$ to be associated to a Hilbert modular form $f$ of weight $k = (1, 1, 2)$. Thus, we might hope to show that $\xi \cong \xi_f$ (automorphy of $\xi$), and use it to deduce automorphy of $\rho_A$ using functoriality. However, there are several obstacles to this. Because the entries of $k$ are not all congruent modulo 2, the automorphic representation associated to $f$ is not algebraic as a representation of $\mathrm{GL}_{2,F}$, and while the adjoint $\operatorname{ad}^0 \xi_f$ is well-defined, $\xi_f$ itself is not. Even positing some form of automorphy of $\xi$, when $K$ is a cubic field, the functoriality of the Asai transfer $\mathrm{GL}_{2,K} \xrightarrow{\boxtimes} \mathrm{GL}_{8,\mathbf{Q}}$ remains open. As such, we need a different approach.

Let $V := H^1(A(\mathbf{C}), \mathbf{Q})$. By work of Galluzzi [Gal00], there is a 9-dimensional weight 2 Hodge substructure $W \subset \operatorname{Sym}^2 V$ of K3-type, and the Kuga–Satake lift $\mathrm{KS}(W)$ is a polarized weight 1 Hodge structure corresponding to an abelian variety isogenous to $A^{32}$. Repeating Galluzzi's arguments on the Galois side, there exists a 9-dimensional representation $\rho \cong \operatorname{Ind}_{G_K}^{G_{\mathbf{Q}}} \operatorname{ad}^0 \xi \subset \operatorname{Sym}^2 \rho_A(1)$ with image landing in $\mathrm{SO}_9(\overline{\mathbf{Q}_\ell})$. Even though $\xi$ is not Hodge–Tate, $\operatorname{ad}^0 \xi$ is Hodge–Tate, and $\rho$ has Hodge–Tate weights $(-1, 0, \dots, 0, 1)$. Our speculative strategy for obtaining an automorphy result is as follows. First, prove automorphy of $\operatorname{ad}^0 \xi$ using the modularity lifting ideas of [BT99, Buz03] and the residual modularity theorems of [AKT21], and deduce automorphy of $\rho$ by functoriality. Second, use the lifting results of Patrikis [Pat19] to lift automorphy of $\rho$ to automorphy of the representation corresponding to the Kuga–Satake lift $\mathrm{KS}(W)$. Third, use the isogeny $\mathrm{KS}(W) \sim A^{32}$ to deduce that $A^{32}$ is automorphic. Fourth, use uniqueness results on the factorization of automorphic $L$-functions to deduce that $A$ itself is automorphic.

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